AMD DEVELOPER HACKATHON ACT II / TRACK 3

One Brain. Multiple Bodies.

Iris is a persistent personal AI that can use the devices a person already owns: desktop GPU, voice, phone location, notifications, and native wrist interaction.

CAPABILITY FABRIC

The difference

Devices do not merely run an app. They announce live capabilities to the Brain. The Brain decides where work and actions should happen.

Hermes is the agent engine. Iris is the persistent multi-device product layer.

A live protocol connects the Brain to the user's world.

GPU + VOICE

Desktop

Local Gemma endpoint + F5-TTS voice.

ACTION + CONTEXT

Android

notification.send and location.current over a Ktor WebSocket.

INTERACTION

Wear OS

Native voice, TTS, sessions, paired-phone sign-in.

PERSISTENT — ONE PER USER

Hermes Brain

Registry + Orchestrator pick live providers, correlate responses, and handle presence and failover.

`announce -> live registry -> route action / inference -> correlated response`

The technical novelty: presence-aware capability routing.

A Body announces what is available. The Registry updates on connect or disconnect. The Orchestrator chooses a provider, sends a correlated request, and recovers between turns.

WIRE FORMAT

```
hermes.capabilities.announce
{ device: 'android',
  notification: { available: true },
  location: { available: true } }

disconnect -> capability leaves registry
```

WIRE FORMAT

```
notification.send.request
{ title, body, request_id }

device.action.response
{ request_id, success, device }

first valid response wins
```

Validated on real Android hardware

A Galaxy S20 FE connected to the production gateway, received a real high-priority notification, and returned a live GPS fix on demand. If no Body is present, the integration returns a clear HTTP 503.

Your GPU first. AMD cloud when needed.

Iris maintains one OpenAI-compatible inference contract while moving execution across available AMD compute routes.

AMD LOCAL

Radeon RX 9060 XT

Gemma runs locally through LM Studio beside Dot's F5-TTS voice. This is the private first route when the desktop is online.

AMD CLOUD

AMD MI300 pod

The current cloud runtime hosts Gemma through vLLM. It is the AMD-hosted route when local capacity is unavailable.

VALIDATION

Radeon PRO W7900

Gemma Q4 was tested in the hackathon environment through llama.cpp, proving the route on AMD workstation hardware too.

FALLBACK

Resilience

Fireworks AI API is wired as the final external fallback route.

`RX 9060 XT / local Gemma -> MI300 / vLLM -> Fireworks API`

Conversation state stays in the Brain; if a route disappears, the next turn advances to the next provider.

A SaaS delivery path already exists behind the demo.

Iris is not a landing page around a prototype: account creation, checkout, automated provisioning, and tenant isolation are part of the product architecture.

01

Account

Django account and customer console.

02

Plan

Stripe checkout for managed compute or BYOK.

03

Provision

Dedicated Ubuntu tenant created automatically.

04

Connect

A private Brain receives its own Bodies.

Per-tenant isolation

Every tenant receives a unique Unix identity, Hermes runtime, memory, credentials, environment, and services. The system has also been used in real multi-tenant deployments.

An agent that persists - and can act through the real world.

Iris turns personal AI from an isolated chat session into an owned Brain with expandable, live capabilities.

CORE INNOVATION

Original

A presence-aware Capability Protocol + Orchestrator: devices announce what they can do; the Brain routes actions and inference across live providers.

WORKING SYSTEM

Technically real

Authenticated gateway, Registry, Android actions, desktop local services, AMD-hosted Gemma, and explicit failover behavior.

PRODUCT PATH

Ready to grow

A user can create an account, pay, receive an isolated Brain, and attach their own devices and compute.

Iris is the capability fabric through which Hermes interacts with the world.

github.com/diegoberan/iris | iris.dberan.dev